

Part VI — Heat, Temperature, and Thermodynamics · Chapter 32

Gases and Pressure

How particle collisions create pressure, and how gas variables govern one another through universal laws

Inflate a bicycle tyre and it becomes rigid, able to support a cyclist. Release the valve and the air rushes out with enough force to spin the wheel. An aerosol can left on a hot dashboard can explode. Mountaineers must breathe supplementary oxygen above 7000 m because the atmosphere thins so much that each lungful contains too few oxygen molecules to sustain the body. A syringe pulled back with the tip sealed draws a partial vacuum, and the barrel resists being pushed in. All of these phenomena come from the same source: the ceaseless, random motion of billions of gas molecules, colliding with every surface they contact. This chapter develops the quantitative relationships between gas pressure, volume, temperature, and amount — relationships that are among the most precisely tested and practically useful in all of physics.

32.1 What Makes Gases Unique

Gases differ from solids and liquids in several fundamental ways, all traceable to the same microscopic fact: in a gas, the average distance between molecules is much larger than the size of the molecules themselves. Intermolecular forces, which dominate in solids and liquids, are negligible for most of the time in a gas. Each molecule moves freely until it collides with another molecule or with the container wall.

Because molecules in a gas are widely separated and interact weakly, gases are highly compressible: the volume can be greatly reduced by applying pressure. Liquids and solids are nearly incompressible by comparison. Gases also expand without limit to fill any container, have very low density, and mix readily with other gases (Dalton's Law of Partial Pressures).

These properties make gases both practically important and theoretically tractable. The ideal gas model — in which molecules are treated as perfectly elastic point particles with no intermolecular forces except at collisions — is one of the most successful theoretical models in physics, accurately predicting the behavior of real gases under most conditions.

Key Point: Gas behavior is simpler than solid or liquid behavior precisely because the molecules interact so little. The gas laws arise from pure mechanics — Newton's laws applied to billions of particles simultaneously. This is why ideal gas predictions are so accurate: the physics is clean.

32.2 Pressure as Force per Unit Area

Pressure is force applied perpendicular to a surface per unit area of that surface. It is the most fundamental macroscopic quantity describing a gas in contact with any surface.

$$P = F / A$$

Pressure P (pascals) = perpendicular force F (newtons) $\div$ area A (m^2). $1 \text{ Pa} = 1 \text{ N/m}^2$.

Pressure is a scalar (it has magnitude but no direction), while force is a vector. The force exerted by a gas on a surface always points perpendicular to and outward from the surface, regardless of orientation. A gas in a container exerts the same pressure in every direction — upward, downward, and sideways equally. This is Pascal's principle.

Unit	Definition	Conversion
Pascal (Pa)	SI unit; 1 N/m^2	Standard atmospheric pressure = 101 325 Pa
kilopascal (kPa)	1000 Pa	$1 \text{ atm} = 101.325 \text{ kPa}$
Atmosphere (atm)	Standard atmosphere	$1 \text{ atm} = 101\,325 \text{ Pa}$
Bar	10^5 Pa	$1 \text{ atm} \approx 1.013 \text{ bar}$
Millimetre Hg (mmHg)	Pressure of 1 mm mercury column	$1 \text{ atm} = 760 \text{ mmHg (torr)}$
Pounds per in ² (psi)	US customary	$1 \text{ atm} = 14.7 \text{ psi}$

Table 32.1 *Pressure units and conversions. The pascal (Pa) is the SI unit.*

Example — Atmospheric pressure on a palm Your palm has area $\approx 0.010 \text{ m}^2$. Atmospheric pressure = 101 325 Pa. Force on palm = $P \times A = 101\,325 \times 0.010 = 1013 \text{ N} \approx 100 \text{ kg-force}$. The air pushes on your palm with the equivalent of a 100 kg weight, but the same pressure acts on the back of your hand in the opposite direction, so you feel no net force.

32.3 Microscopic Origin of Gas Pressure

Gas pressure is caused by the continuous bombardment of surfaces by moving gas molecules. Each collision transfers a tiny impulse to the wall. Because a typical gas at atmospheric pressure contains about 3×10^{25} molecules per cubic meter, the number of collisions per unit area per second is enormous — typically 10^{23} per cm^2 per second. Each collision is imperceptibly small, but together they produce a steady, macroscopic force on every surface.

From the kinetic theory analysis, the pressure exerted by an ideal gas of N molecules in volume V is:

$$P = (1/3) (N/V) m v_{\text{rms}}^2$$

Kinetic theory pressure. N = number of molecules; V = volume; m = mass per molecule; v_{rms} = root-mean-square speed. Shows P depends on molecular KE and number density.

Since the average translational KE is $\frac{1}{2}mv_{\text{rms}}^2 = (3/2)k_{\text{B}} T$ (from Chapter 29), the pressure can be written as $P = Nk_{\text{B}} T / V$, which rearranges to $PV = Nk_{\text{B}} T$ — the microscopic form of the ideal gas law. Pressure emerges from molecular kinetics; it is not a separate property but a macroscopic manifestation of particle collisions.

This microscopic view immediately explains the effects of changing P , V , and T . Increasing temperature increases molecular speeds, so collisions are more energetic and more frequent — pressure rises. Decreasing volume means molecules travel shorter distances between wall collisions, so they hit more frequently — pressure rises. Adding more molecules (increasing N) at fixed V and T means more collisions per second — pressure rises.

32.4 Gas Variables: Pressure, Volume, Temperature, Amount

The macroscopic state of a fixed amount of ideal gas in equilibrium is completely determined by three variables: pressure P , volume V , and absolute temperature T . A fourth quantity, the amount of gas (n moles or N molecules), scales the other three.

The amount of gas is measured in moles. One mole contains Avogadro's number of molecules: $N_{\text{A}} = 6.022 \times 10^{23}$ particles/mol. The number of moles $n = (\text{total mass}) / (\text{molar mass})$. For oxygen (O_2 , molar mass 32 g/mol), 64 g of oxygen is 2.0 mol; at STP this occupies $2.0 \times 22.4 = 44.8$ L.

Standard Temperature and Pressure (STP) is defined as $T = 273.15$ K (0°C) and $P = 100$ kPa (1 bar, by modern IUPAC definition; older texts use 101.325 kPa). At STP, one mole

of ideal gas occupies 22.4 L. This molar volume is a useful reference for gas calculations.

The Rule: Always use absolute temperature (kelvins) in all gas law equations. Using Celsius temperatures in P, V, T relations gives completely wrong results. Convert $T_{\text{C}} + 273.15 = T_{\text{K}}$ before applying any gas law.

32.5 Boyle's Law

Boyle's Law describes the relationship between pressure and volume for a fixed amount of gas at constant temperature. At constant T and n, pressure and volume are inversely proportional:

$$PV = \text{constant}$$

 $\Rightarrow$

$$P_1V_1 = P_2V_2$$

Boyle's Law. T and n constant. Doubling the pressure halves the volume; halving the pressure doubles the volume.

Boyle's Law reflects the microscopic picture directly: if you halve the volume at constant temperature, the molecular speed distribution does not change (temperature is the same), but the molecules now hit the walls twice as often (shorter average distance between collisions). So pressure doubles.

Example — Boyle's Law calculation A gas at 2.0 atm occupies 3.0 L. It expands to 5.0 L at constant temperature. New pressure: $P_2 = P_1V_1/V_2 = (2.0 \times 3.0)/5.0 = 1.2$ atm. The pressure dropped because the molecules now have a larger volume to fill.

Historical Note: Robert Boyle published his pressure-volume relationship in 1662 in 'New Experiments Physico-Mechanical.' His apparatus used a J-shaped glass tube with trapped air; he systematically measured pressure and volume by adding mercury to the tube. The relationship $P \propto 1/V$ at constant temperature was the first quantitative gas law — and one of the earliest examples of a quantitative physical law derived from careful experiment.

32.6 Charles's Law

Charles's Law describes the relationship between volume and temperature for a fixed amount of gas at constant pressure. At constant P and n, volume is proportional to absolute temperature:

$$V/T = \text{constant}$$

 $\Rightarrow$

$$V_1/T_1 = V_2/T_2$$

Charles's Law. P and n constant. T must be in kelvins. Doubling T (in K) doubles V.

Microscopically: at constant pressure, the walls of a flexible container adjust until the pressure inside equals the outside. Higher temperature means faster molecules hitting harder and more often; the container must expand until the pressure (hits per area per time) returns to equilibrium. Volume and temperature are proportional.

Charles's Law explains many everyday observations: a sealed bag of chips appears to expand at altitude (lower external pressure and higher cabin temperature both contribute); a car tyre is slightly under-inflated in cold weather compared to what was put in on a warm day; balloons deflate when taken outside on a cold day.

Example — Charles's Law calculation A gas balloon has volume 2.0 m^3 at 20°C (293 K). It rises to where $T = -30^\circ\text{C}$ (243 K) at constant pressure. New volume: $V_2 = V_1 T_2/T_1 = 2.0 \times 243/293 = 1.66 \text{ m}^3$. The balloon shrinks by about 17%.

32.7 Gay-Lussac's Law

Gay-Lussac's Law (also called Amonton's Law) describes the relationship between pressure and temperature for a fixed amount of gas at constant volume. At constant V and n , pressure is proportional to absolute temperature:

$$P/T = \text{constant} \quad \Rightarrow \quad P_1/T_1 = P_2/T_2$$

Gay-Lussac's Law. V and n constant. T must be in kelvins. Doubling T (in K) doubles P .

This is the most dangerous gas law for engineering. A sealed rigid container (fixed V) that is heated undergoes pressure rise proportional to absolute temperature. An aerosol can at 20°C (293 K) has a design pressure. If heated to 100°C (373 K), the pressure rises by a factor of $373/293 = 1.27$, a 27% increase. At 250°C (523 K), the pressure nearly doubles. This is why aerosol cans and pressurised cylinders must never be heated or incinerated.

Example — Pressure in a heated tyre A car tyre has gauge pressure 2.2 atm above atmospheric (absolute pressure 3.2 atm) at 15°C (288 K). After driving, the tyre temperature rises to 55°C (328 K). New absolute pressure: $P_2 = 3.2 \times 328/288 = 3.64 \text{ atm}$. Gauge pressure = $3.64 - 1 = 2.64 \text{ atm}$. A rise of about 0.44 atm, which is why manufacturers specify cold-tyre inflation pressure.

32.8 Combined Gas Law

The three individual gas laws can be combined into a single relationship for a fixed amount of gas (constant n) undergoing any process that changes two or all three of P , V , and T simultaneously:

$$PV/T = \text{constant} \quad \Rightarrow \quad P_1V_1/T_1 = P_2V_2/T_2$$

Combined Gas Law. n constant; T in kelvins. The individual laws are special cases: Boyle (T constant), Charles (P constant), Gay-Lussac (V constant).

The combined gas law is the workhorse of gas calculations: most real problems involve simultaneous changes in at least two of P , V , T . The procedure is always the same: identify the initial and final states, list all six values (P_1 , V_1 , T_1 , P_2 , V_2 , T_2) with one unknown, convert temperatures to kelvins, and solve.

Step 1: Convert: All temperatures to Kelvin. Pressures must be in the same units (either all absolute or all gauge; use absolute for thermodynamic relationships).

Step 2: Identify: Which variables are constant (use the simpler individual law if two are constant).

Step 3: Apply: $P_1V_1/T_1 = P_2V_2/T_2$ and solve for the unknown.

Step 4: Check: Are units consistent? Does the answer make physical sense (larger $T \rightarrow$ larger P or V)?

Example — Combined gas law A gas at $P_1 = 1.5$ atm, $V_1 = 4.0$ L, $T_1 = 300$ K is changed to $T_2 = 400$ K and $P_2 = 2.0$ atm. Find V_2 . $V_2 = P_1V_1T_2/(T_1P_2) = (1.5 \times 4.0 \times 400)/(300 \times 2.0) = 2400/600 = 4.0$ L. The volume is unchanged because the increased temperature effect (would increase V) is exactly cancelled by the increased pressure (would decrease V).

32.9 Ideal Gas Law (Introductory)

The ideal gas law combines all four gas variables — pressure, volume, temperature, and amount — into a single equation valid for any state of an ideal gas (not just comparing two states):

$$PV = nRT$$

Ideal Gas Law. P = pressure (Pa); V = volume (m^3); n = amount (mol); $R = 8.314$ J/(mol·K) (universal gas constant); T = absolute temperature (K).

The ideal gas law is a single equation of state: given any three of P , V , n , T , you can find the fourth. It is valid for real gases at low pressures and high temperatures (conditions where the molecular volume and intermolecular forces are negligible).

An equivalent form using the number of molecules N (rather than moles n): $PV = Nk_B T$, where $k_B = R/N_A = 1.381 \times 10^{-23}$ J/K. The two forms are related by $N = nN_A$.

Example — Finding density from ideal gas law At $T = 300$ K and $P = 101\,325$ Pa, find the density of nitrogen gas ($M = 28.0$ g/mol = 0.0280 kg/mol). $n/V = P/(RT) =$

$101\,325/(8.314 \times 300) = 40.6 \text{ mol/m}^3$. Density = $(n/V) \times M = 40.6 \times 0.0280 = 1.14 \text{ kg/m}^3$. Compare to air at STP: 1.293 kg/m^3 (air is mostly N_2 , so this is reasonable).

Example — Ideal gas law calculation 2.0 mol of helium at 25°C (298 K). Find volume at $P = 200 \text{ kPa} = 200\,000 \text{ Pa}$. $V = nRT/P = (2.0 \times 8.314 \times 298)/200\,000 = 4955/200\,000 = 0.0248 \text{ m}^3 = 24.8 \text{ L}$.

32.10 Real Gases vs. Ideal Gases

The ideal gas law assumes (1) molecules have negligible volume compared to the container, and (2) intermolecular forces are negligible except during elastic collisions. These assumptions fail at high pressures (molecules are forced close together, volume and forces matter) and low temperatures (molecules move slowly and intermolecular attraction becomes significant, causing clustering).

Van der Waals corrected the ideal gas equation to account for both effects. The Van der Waals equation is not required at this level, but it is important to know when ideal gas approximations break down: above a few atmospheres of pressure, or near the condensation (liquefaction) temperature of the gas, real gas behavior deviates measurably from ideal.

Most gases at room temperature and atmospheric pressure behave very nearly ideally. The ideal gas law predicts the properties of air, nitrogen, oxygen, argon, and helium to within fractions of a percent under ordinary conditions. Deviations become significant for gases like CO_2 , NH_3 , and H_2O at high pressures or low temperatures.

The compressibility factor $Z = PV/(nRT)$ measures deviation from ideal behavior. For an ideal gas, $Z = 1$ exactly. Real gases have $Z > 1$ at high pressures (repulsive molecular volume dominates) and $Z < 1$ at intermediate pressures (attractive intermolecular forces dominate). At sufficiently high pressure, all real gases have $Z > 1$.

32.11 Pressure Measurement and Instruments

Pressure is measured by a variety of instruments depending on the range, accuracy, and type of measurement required.

A manometer is a U-shaped tube partially filled with a liquid (usually mercury or water). One side is connected to the gas being measured; the other is open to atmosphere. The height difference h between the two liquid columns is proportional to the gauge pressure (pressure difference from atmospheric): $P_{\text{gauge}} = \rho gh$.

A barometer measures atmospheric pressure. The classic mercury barometer (Torricelli,

1643) is a sealed mercury tube inverted into a mercury reservoir. Atmospheric pressure supports a mercury column of height $h \approx 760 \text{ mm}$ at sea level. Atmospheric pressure in Pa: $P_{\text{atm}} = \rho_{\text{Hg}} g h = 13\,600 \times 9.81 \times 0.760 = 101\,300 \text{ Pa}$.

Gauge pressure is pressure above atmospheric. Absolute pressure is gauge pressure plus atmospheric: $P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}$. A flat tyre has gauge pressure zero ($P_{\text{abs}} = P_{\text{atm}}$). A completely evacuated container has absolute pressure zero (perfect vacuum) and gauge pressure $-P_{\text{atm}}$. Always use absolute pressure in gas law calculations.

Study Tip: In gas law problems: always use absolute pressure. If given gauge pressure, add 101.325 kPa (or 1 atm, or 14.7 psi) to get absolute. Always use Kelvin. Mixed units (kPa with atm, or Celsius with kelvins) cause large errors.

32.12 Everyday and Engineering Applications of Gas Laws

Gas laws govern an enormous range of practical phenomena and engineering systems.

Internal combustion engines

The engine cycle compresses a fuel-air mixture (Boyle's Law at approximately constant temperature), ignites it (Gay-Lussac: pressure rises at constant volume), allows the hot gas to expand and do work on the piston (combined gas law), and exhausts the spent gas. The efficiency of the engine is directly related to the compression ratio — how much the gas is compressed before ignition.

Breathing and lung mechanics

The diaphragm contracts and flattens to increase lung volume (Boyle's Law: volume increases $\rightarrow$ pressure drops below atmospheric $\rightarrow$ air flows in). It relaxes to decrease volume (pressure rises above atmospheric $\rightarrow$ air flows out). The driving force for breathing is a pressure difference of only about 2–3 mmHg relative to atmospheric.

Altitude and aviation

At 10 000 m altitude, atmospheric pressure is about 26 kPa — only 26% of sea-level pressure. Cabin pressurisation in aircraft maintains cabin pressure at about 75 kPa (equivalent to 2500 m altitude). The fuselage is a pressure vessel, and the pressure differential across its walls requires strong, lightweight construction.

Weather and barometric pressure

Atmospheric pressure varies with altitude, temperature, and weather systems. High-pressure systems (anticyclones) bring clear weather; low-pressure systems (cyclones) bring storms. A barometer reading rising rapidly indicates improving weather; falling

rapidly indicates approaching storm. The gas law relationship $PV = nRT$, applied to the atmosphere, underpins all quantitative weather modelling.

SCUBA diving

At 10 m depth, water pressure is about 2 atm absolute. A SCUBA diver breathes air at ambient pressure. If a diver holds their breath while ascending (violating Boyle's Law deliberately), lung volume expands and can rupture — a potentially fatal consequence of ignoring the gas law. SCUBA training emphasises: always exhale during ascent.

32.13 Common Mistakes in Gas Law Problems

- Using Celsius instead of Kelvin. The most common and most consequential error. All gas law equations require absolute temperature. Always convert: $T_K = T_C + 273.15$.
- Using gauge pressure instead of absolute pressure. Gas laws require absolute pressure. If a problem gives tyre pressure as 2.2 atm gauge, the absolute pressure is $2.2 + 1.0 = 3.2$ atm.
- Applying the combined gas law when n is not constant. Boyle's, Charles's, Gay-Lussac's, and the combined gas law all assume fixed n . If gas escapes or is added, use the ideal gas law directly with explicit n .
- Misidentifying constant variables. When one variable is explicitly constant (e.g., a rigid container $\rightarrow V$ constant; open container $\rightarrow P$ constant), use the simpler individual law. Students sometimes substitute $T = \text{constant}$ into the combined law and leave T_2 unknown.
- Inconsistent units. Both states in $P_1V_1/T_1 = P_2V_2/T_2$ must use the same units for each variable (P in atm both times, V in litres both times, T in kelvins both times). Mixing atm and Pa or litres and m^3 gives wrong answers.
- Forgetting to use $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$ with SI units in $PV = nRT$. If P is in Pa and V is in m^3 , then PV is in $\text{Pa}\cdot\text{m}^3 = \text{J} = \text{N}\cdot\text{m}$, and $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$ is correct. If P is in kPa or atm or V in litres, use the appropriate value of R or convert to SI first.

Common Mistake: Always: (1) convert T to Kelvin, (2) use absolute pressure, (3) ensure consistent units on both sides. These three checks catch 90% of gas law errors.

Chapter Summary

Gas pressure arises from the continuous bombardment of surfaces by rapidly moving molecules. The macroscopic gas laws emerge from this microscopic kinetics.

- Boyle's Law (T, n constant): $P_1V_1 = P_2V_2$. Pressure and volume inversely proportional.
- Charles's Law (P, n constant): $V_1/T_1 = V_2/T_2$. Volume and absolute temperature proportional.
- Gay-Lussac's Law (V, n constant): $P_1/T_1 = P_2/T_2$. Pressure and absolute temperature proportional.
- Combined Gas Law (n constant): $P_1V_1/T_1 = P_2V_2/T_2$. General two-state comparison.
- Ideal Gas Law (any state): $PV = nRT$. $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$. T in kelvins; P in pascals; V in m^3 .
- Real gases deviate from ideal at high pressures and low temperatures (near condensation).
- Always use absolute pressure and Kelvin in all gas equations.

Key Terms

pressure (P): Force per unit area perpendicular to a surface; $P = F/A$; SI unit: pascal (Pa = N/m^2).

ideal gas: A theoretical gas model in which molecules have negligible volume and no intermolecular forces except perfectly elastic collisions; obeys $PV = nRT$ exactly.

mole (mol): The SI unit of amount of substance; $1 \text{ mol} = 6.022 \times 10^{23}$ particles (Avogadro's number N_A).

universal gas constant (R): $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; the proportionality constant in the ideal gas law $PV = nRT$.

Boyle's Law: At constant T and n , $PV = \text{constant}$; pressure and volume are inversely proportional.

Charles's Law: At constant P and n , $V/T = \text{constant}$; volume and absolute temperature are proportional.

Gay-Lussac's Law: At constant V and n , $P/T = \text{constant}$; pressure and absolute temperature are proportional.

combined gas law: $P_1V_1/T_1 = P_2V_2/T_2$; applies when n is constant and two or three of P , V , T change.

ideal gas law: $PV = nRT$; the equation of state for an ideal gas relating all four gas variables in a single equation.

STP (Standard Temperature and Pressure): $T = 273.15 \text{ K}$ (0°C) and $P = 100 \text{ kPa}$; at STP, 1 mol of ideal gas occupies 22.4 L.

absolute pressure: Total gas pressure measured from zero (perfect vacuum); $P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}$.

gauge pressure: Pressure relative to atmospheric pressure; what most pressure gauges measure.

compressibility factor (Z): $Z = PV/(nRT)$; measures deviation from ideal gas behavior. $Z = 1$ for a perfect ideal gas.

Van der Waals equation: An equation of state correcting the ideal gas law for molecular volume and intermolecular attraction; applies to real gases at high pressures.

Concept Review Questions

Answer in complete sentences. No calculations required unless specified.

1. Explain, using the microscopic picture, why gas pressure increases when the temperature is raised at constant volume.
 2. State Boyle's Law in words and explain it microscopically. Why does halving the volume double the pressure at constant temperature?
 3. Why must temperature be expressed in Kelvin, not Celsius, in Charles's Law? What goes wrong numerically if you use 0°C instead of 273 K?
 4. A sealed aerosol can is accidentally placed in a fire. Explain the danger using Gay-Lussac's Law.
 5. A diver's lung volume at 20 m depth (3 atm absolute) is 4.0 L. If the diver holds their breath and ascends to the surface (1 atm), what is the new lung volume? Why is this dangerous?
 6. Explain the difference between absolute pressure and gauge pressure. Give two examples where confusion between them would lead to serious error.
 7. At what conditions does real gas behavior deviate significantly from ideal gas behavior? Explain the molecular reasons for each type of deviation.
 8. Why does air pressure decrease with increasing altitude? How does this affect the boiling point of water at high altitude?
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Pressure Problems

9. A 70 kg person stands on one foot. The area of the foot in contact with the ground is 0.014 m^2 . Find the pressure exerted on the ground in Pa and in kPa.
 10. A scuba tank has gauge pressure 200 atm. What is the absolute pressure in atm and in Pa?
 11. A mercury barometer reads 742 mmHg. Convert to Pa, kPa, and atm. ($\rho_{\text{Hg}} = 13\,600 \text{ kg/m}^3$)
 12. A hydraulic press has a small piston (area 5.0 cm^2) and a large piston (area 200 cm^2). A force of 100 N is applied to the small piston. What force does the large piston exert? (Pressure is transmitted equally: Pascal's principle.)
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Gas Law Calculations

13. A gas at 1.0 atm and 3.0 L is compressed to 0.60 L at constant temperature. Find the new pressure. (Boyle's Law)
 14. A gas occupies 4.0 L at 27°C and constant pressure. Find its volume at 127°C . (Charles's Law)
 15. A gas in a rigid container has pressure 1.5 atm at 20°C . It is heated to 80°C . Find the new pressure. (Gay-Lussac's Law)
 16. Use the ideal gas law to find the volume of 3.0 mol of an ideal gas at $T = 350 \text{ K}$ and $P = 150 \text{ kPa}$.
 17. Find the number of moles of gas in a 10.0 L container at 25°C and 2.50 atm ($1 \text{ atm} = 101\,325 \text{ Pa}$).
 18. Calculate the pressure exerted by 0.50 mol of helium in a 1.0 L container at 300 K.
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Combined Gas Law Exercises

19. A gas at 3.0 atm, 2.0 L, and 300 K changes to 1.5 atm and 450 K. Find the new volume.
20. A weather balloon is launched with $V = 2.0 \text{ m}^3$, $P = 101 \text{ kPa}$, $T = 15^\circ\text{C}$. At altitude, $T = -50^\circ\text{C}$ and $P = 30 \text{ kPa}$. Find the new volume.

21. A gas cylinder at 25°C and gauge pressure 150 atm (absolute: 151 atm) is cooled to −40°C. Find the new absolute pressure and gauge pressure. (V and n constant)
22. A scuba tank holds 12 L of compressed air at 200 atm gauge pressure and 20°C. The diver breathes from it at the surface (1 atm, 20°C). How many litres of breathing-pressure air does the tank hold? (Boyle's Law: same T)

Applied Thermodynamic Problems

23. A car tyre has volume 25 L and is inflated to gauge pressure 2.2 atm at 18°C (291 K). After driving, the tyre warms to 52°C (325 K). Assuming volume is constant, find the new absolute and gauge pressures.
24. A sealed 500 mL glass bottle of soda contains CO₂ at 3.0 atm absolute and 5°C. The bottle is left in a hot car at 55°C. Assuming rigid bottle, find the new pressure. Is it safe? (Typical glass bottle burst pressure $\approx 4\text{--}5$ atm.)
25. A breath of air has volume 0.500 L at 37°C (310 K) and 1.013 atm. Find the number of moles and the number of molecules.
26. The atmosphere at sea level has $P = 101.3$ kPa, $T = 293$ K, and is about 78% N₂ ($M = 28$ g/mol). Find the mass of nitrogen in a room 5 m $\times$ 4 m $\times$ 2.5 m. (Use $PV = nRT$ with $P_{\text{N}_2} = 0.78 \times 101\,300$ Pa.)

Chapter 32 Checkpoint

Before moving on to Chapter 33, confirm you can do each of the following.

I can...	See Section
Explain the microscopic origin of gas pressure using kinetic theory	32.3
Define absolute vs. gauge pressure and convert between them	32.11
Apply Boyle's, Charles's, and Gay-Lussac's Laws with correct units	32.5–32.7
Use the combined gas law for two-state problems at constant n	32.8

Apply $PV = nRT$ to find any one of P , V , n , T given the other three	32.9
State when real gases deviate from ideal behavior	32.10
Always convert to Kelvin and absolute pressure before substituting	32.4–32.9
Identify common errors in gas law calculations and avoid them	32.13

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